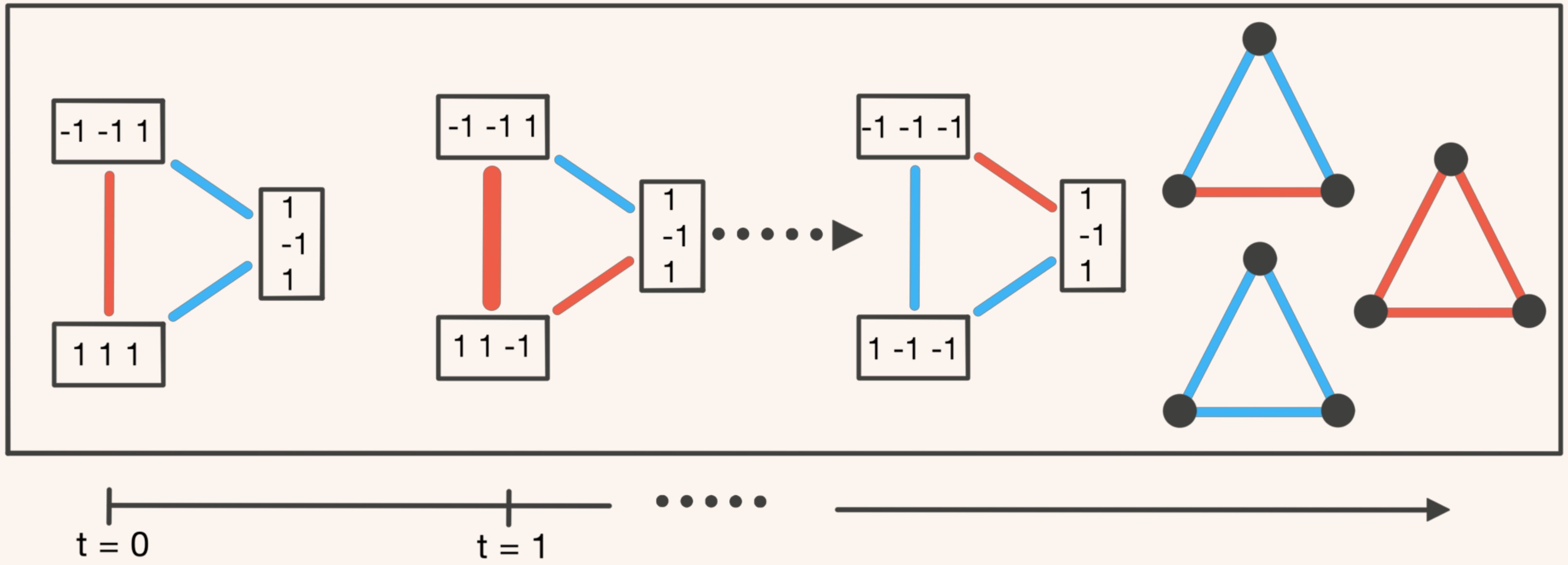
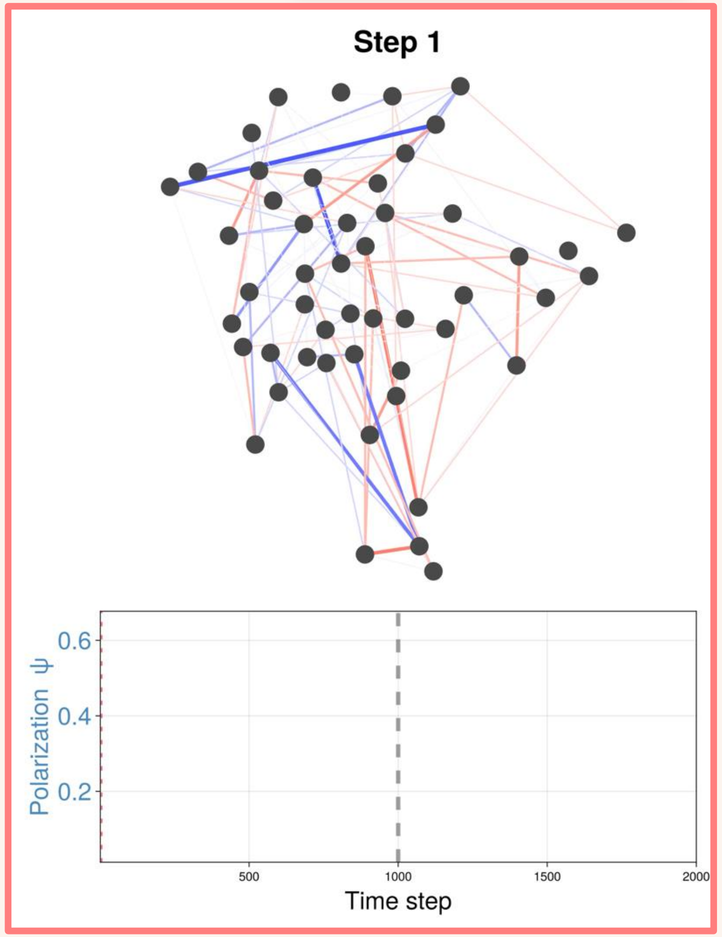
Heterophily as a generative mechanism for self-organized synergistic interdependencies

Enrico Caprioglio
Department of Informatics, University of Sussex





Synergistic high-order interdependencies

statistical interdependencies irreducible to low-order (pairwise) statistics

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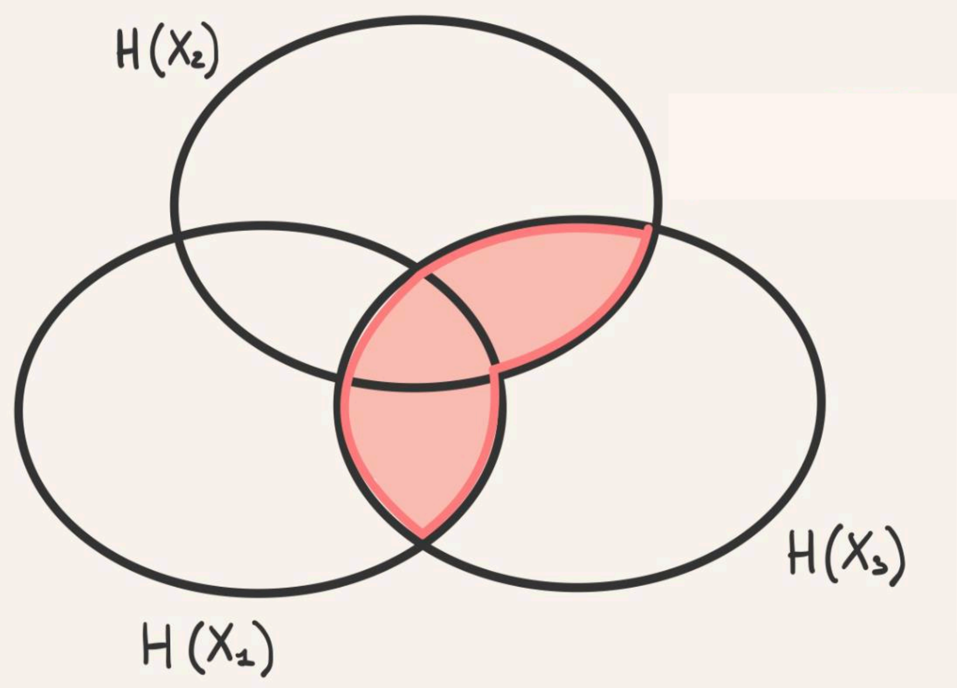
$I(X_3; X_2, X_1)$



Entropy of X: $H(X)$
Mutual Info: $\mathrm{I}(\mathrm{X}; \mathrm{Y})$

Williams & Beer "Nonnegative Decomposition of Multivariate Information"

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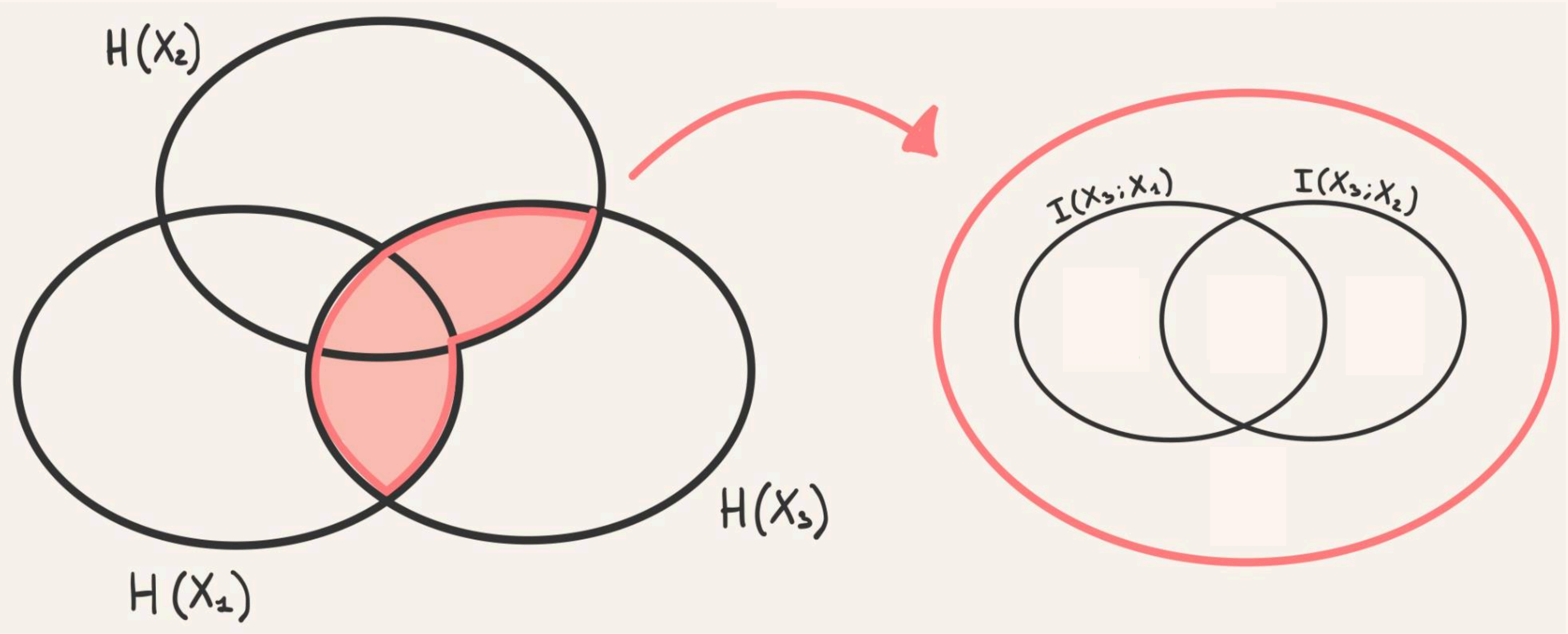

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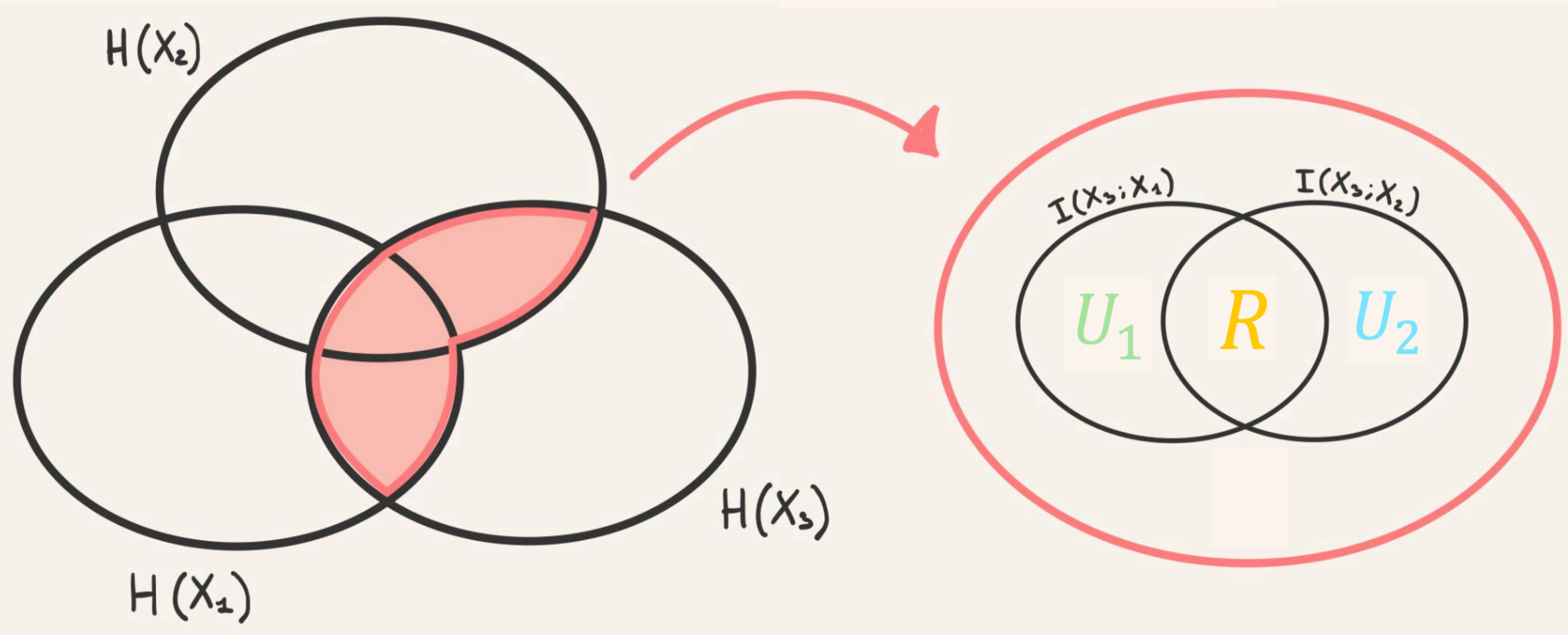

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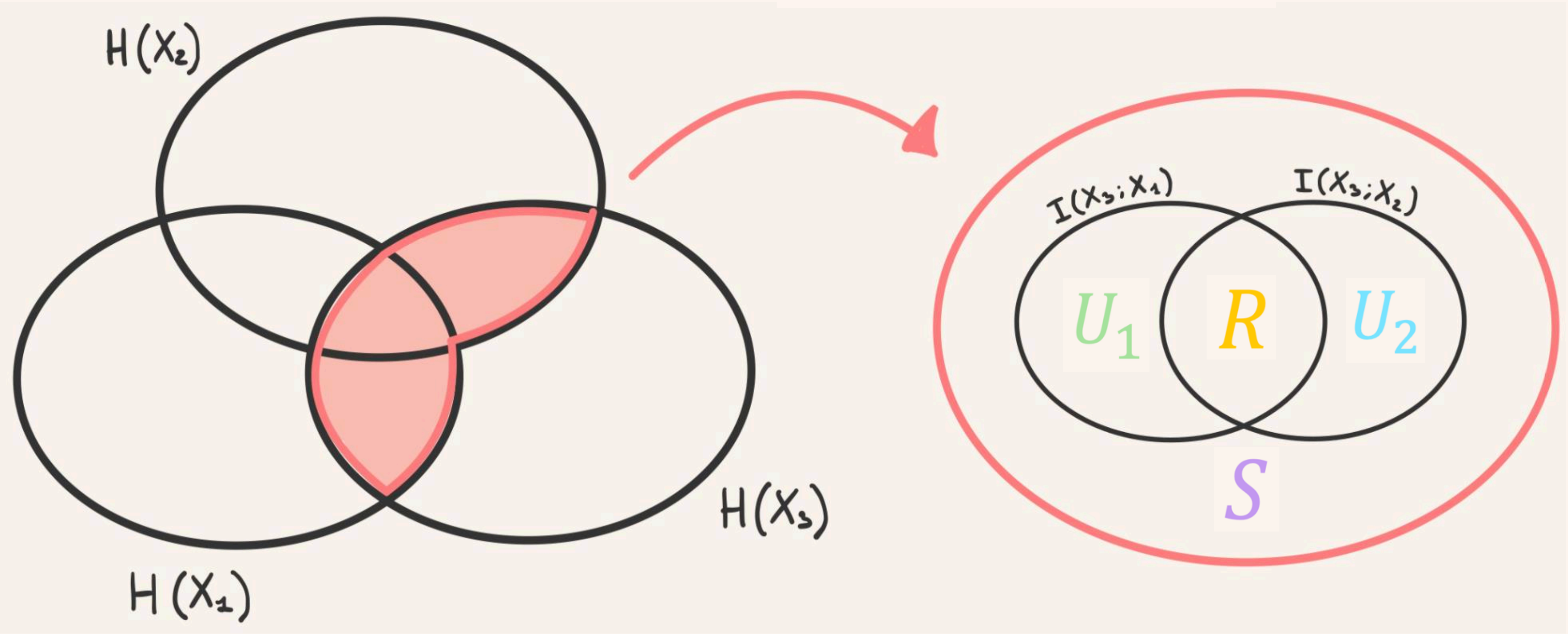

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Synergistic high-order interdependencies

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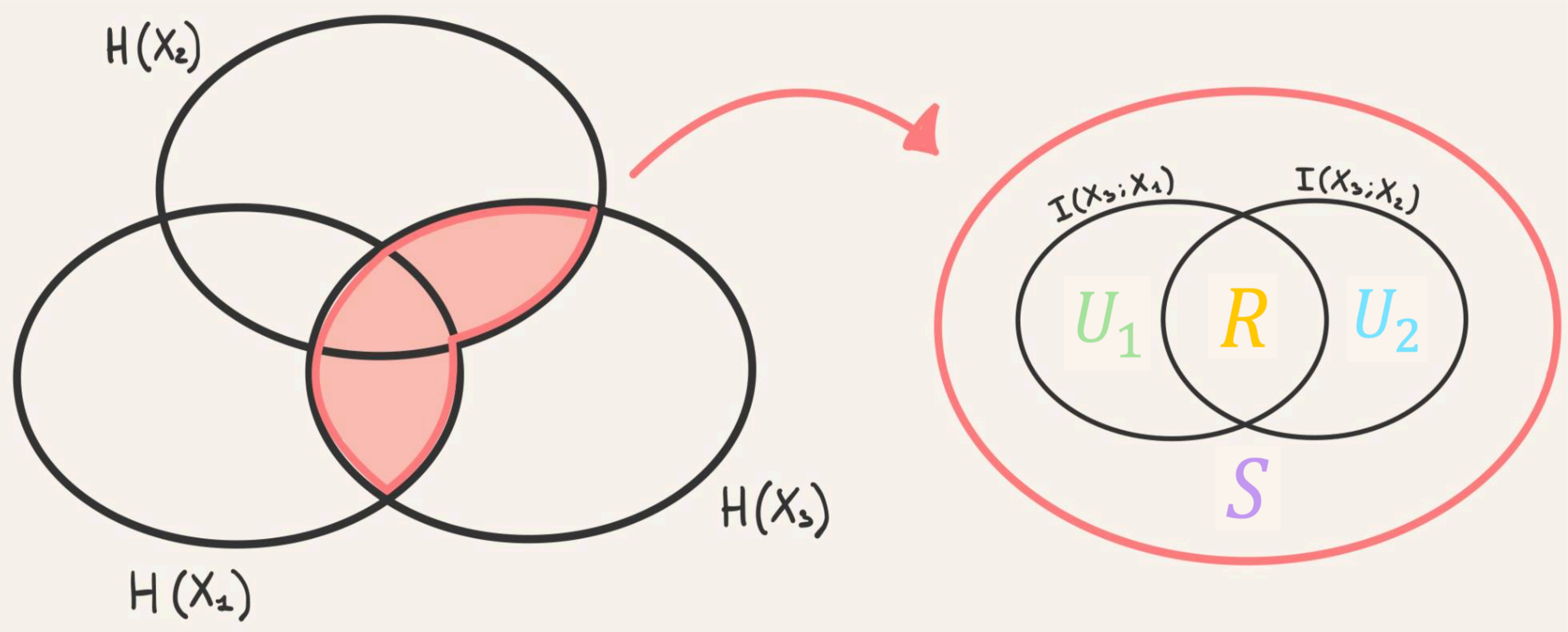
$$S = I(X_3; X_2, X_1) - [U_1 + U_2 + R]$$



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Williams & Beer "Nonnegative Decomposition of Multivariate Information"

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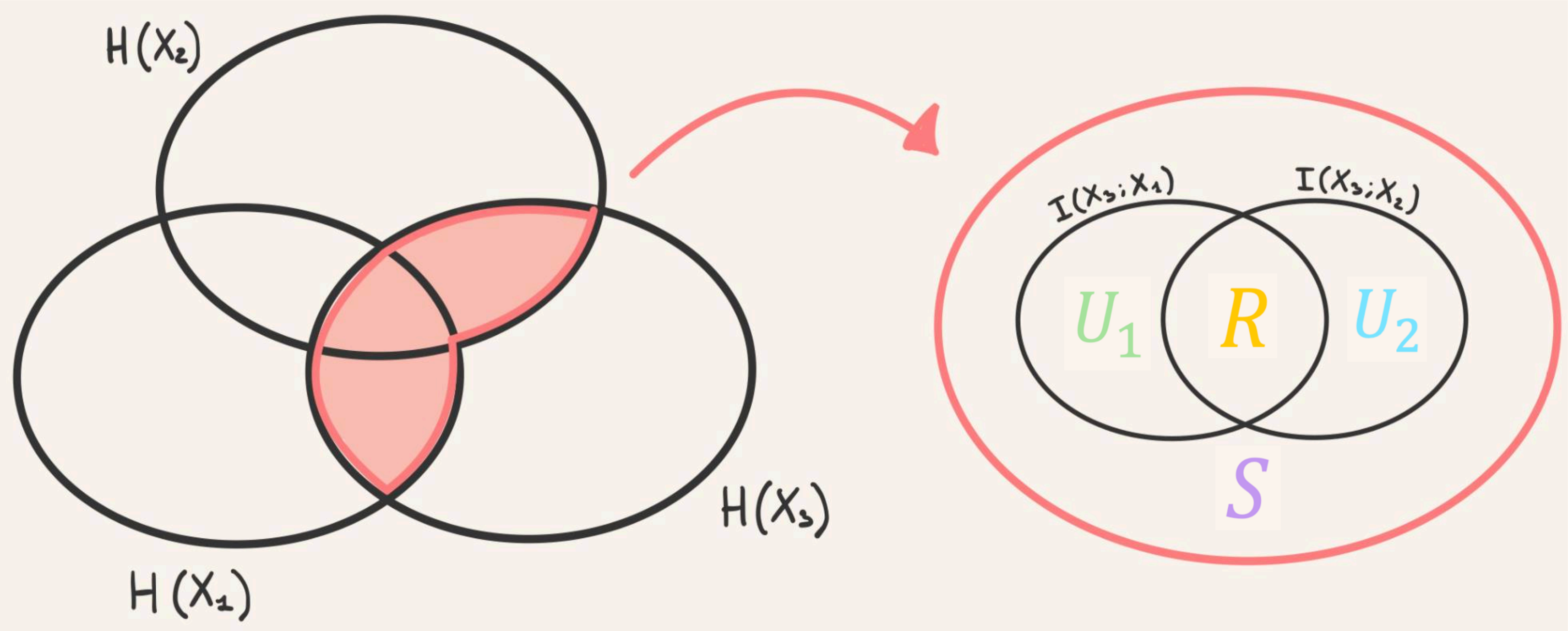


Entropy of X: $H(X)$
Mutual Info: $\mathrm{I}(\mathrm{X};\mathrm{Y})$

Informally: (but enough for this presentation to not just show you equations)

synergies tend to arise when high-order dependencies → e.g., $I(X_3;\ X_2 \mid X_1)$,

are greater than low-order ones → e.g., $I(X_3; X_2)$

Williams & Beer “Nonnegative Decomposition of Multivariate Information”

Synergistic high-order interdependencies

What mechanisms generate collective phenomena?

Known mechanisms, see

- Robiglio et al. “Synergistic Signatures of Group Mechanisms in Higher-Order Systems” Phys. Rev. Lett. (2025)
- Caprioglio et al. “Synergistic Motifs in Gaussian Systems” Phys. Rev. Lett. (2026)

did not account for adaptivity: interactions change in time.

Mechanisms definition: the entities and activities, as well as their organization, to produce the phenomenon of interest.

Luppi et al “A synergistic core for human brain evolution and cognition”, Nature Neuroscience
Varley et al “Emergence of a synergistic scaffold in the brains of human infants”, Comm. Biology
Gatica et al. High-Order Interdependencies in the Aging Brain. Brain Connectivity
Urbina-Rodriguez et al “A Brain-like Synergistic Core in LLMs Drives Behaviour and Learning” ArXiv

Question: what adaptive mechanisms support or give rise to synergistic interdependencies? And how do they do it?

Homophily and heterophily (attraction repulsion form)

Homophily: **Attraction to the similar and repulsion from the dissimilar**

Heterophily: **Attraction to the dissimilar and repulsion from the similar**

Entities: s_i

Similar
Dissimilar
Interactions:

Activities $\rightarrow \ \mathrm{E}^{(i)}$

Homophily and heterophily (attraction repulsion form)

Homophily: **Attraction to the similar and repulsion from the dissimilar**

Heterophily: **Attraction to the dissimilar and repulsion from the similar**

Entities: $\boldsymbol{s}_i = \left[s_i^1, s_i^2, s_i^3, \ldots, s_i^G\right], \quad s_i^g \in [\pm 1]$

Similar if $\boldsymbol{s}_i \cdot \boldsymbol{s}_j > 0$

Dissimilar if: $\boldsymbol{s}_i \cdot \boldsymbol{s}_j < 0$

Interactions: $J_{ij} = A_{ij} \cdot \mathrm{sign}\left(s_i \cdot s_j\right) \in [\pm 1]$

Homophilous entities: $\lambda_i = -1$

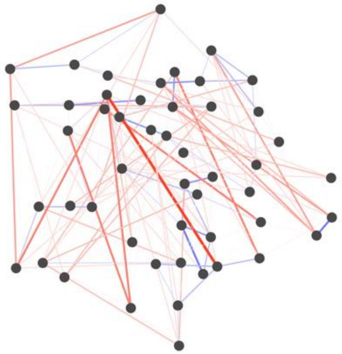
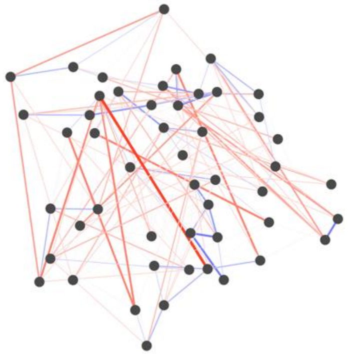
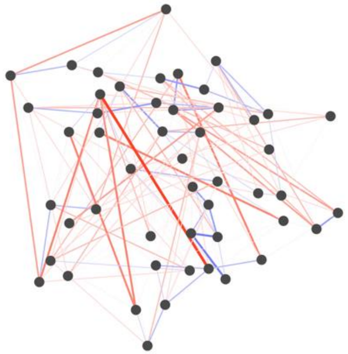
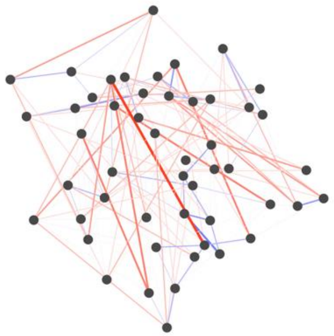
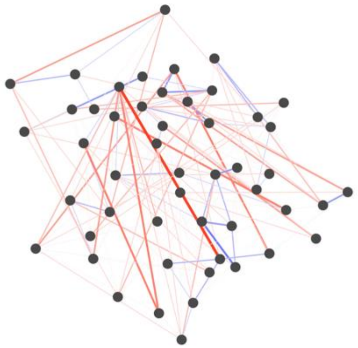
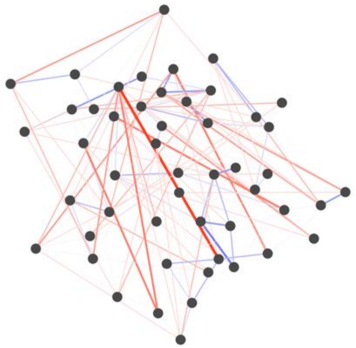
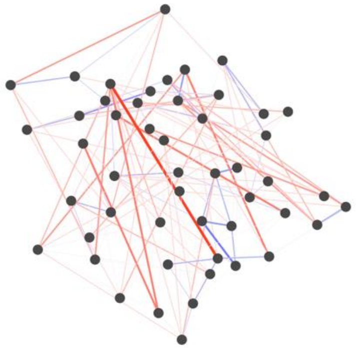
Heterphilous entities: $\lambda_i = 1$

Activities → $\mathrm{E}^{(i)} = \lambda_i \left[\underbrace{\sum_{j:J_{ij}>0} \frac{\alpha}{G} \left(\boldsymbol{s}_i \cdot \boldsymbol{s}_j\right)}_{\text{Positive sum: interactions with the similar}} + \underbrace{\sum_{j:J_{ij}<0} \frac{1-\alpha}{G} \left(\boldsymbol{s}_i \cdot \boldsymbol{s}_j\right)}_{\text{Negative sum: interactions with the dissimilar}} \right]$

Korbel et al “Homophily-based social group formation in a spin glass self-assembly framework”, Physical Review Letters
Pham et al “Empirical social triad statistics can be explained with dyadic homophylic interactions” PNAS

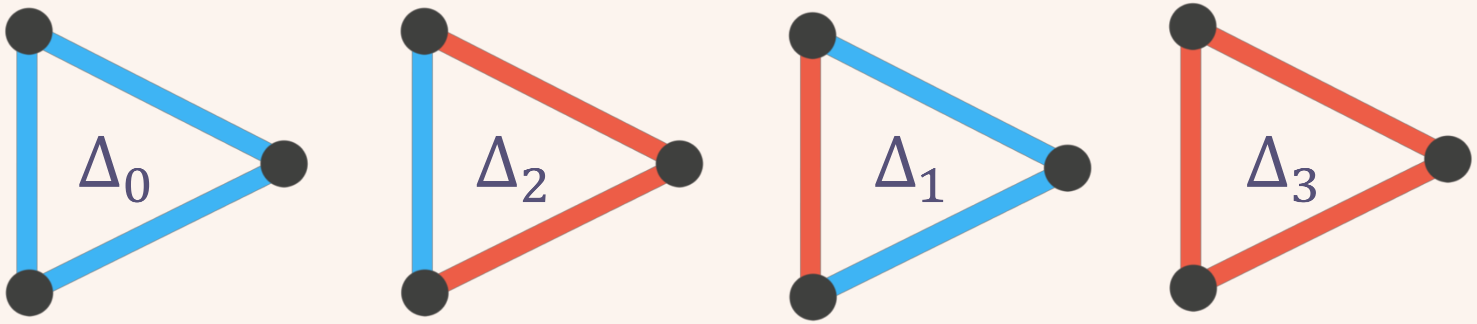
Observation 1:

$$\mathrm{E}^{(i)} = \lambda_i \left[\sum_{j:J_{ij}>0} \frac{\alpha}{\mathrm{G}} (\boldsymbol{s}_i \cdot \boldsymbol{s}_j) + \sum_{j:J_{ij}<0} \frac{1-\alpha}{\mathrm{G}} (\boldsymbol{s}_i \cdot \boldsymbol{s}_j) \right]$$

	$\alpha < 0.5$	$\alpha = 0.5$	$\alpha > 0.5$	
Homophily maximizes similarities and dissimilarities				Random system ($\beta \to 0$) 
Heterophily minimizes similarities and dissimilarities				

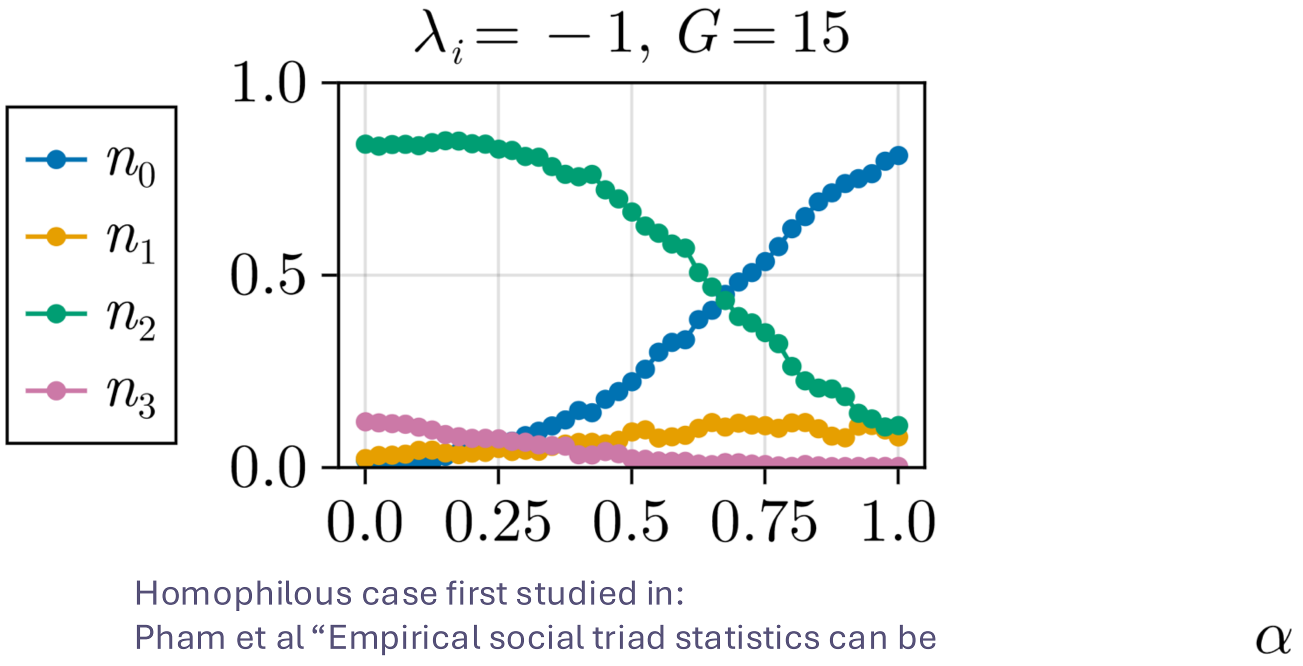
Blue (**Red**) edges indicate **positive** (**negative**) interactions. Edge thickness proportional to $|\boldsymbol{s}_i \cdot \boldsymbol{s}_j|$

Observation 2: macroscopic patterns emerge



Δ_k: with k being # neg. edges

n_k: density of Δ_k triangles



Homophilous case first studied in:
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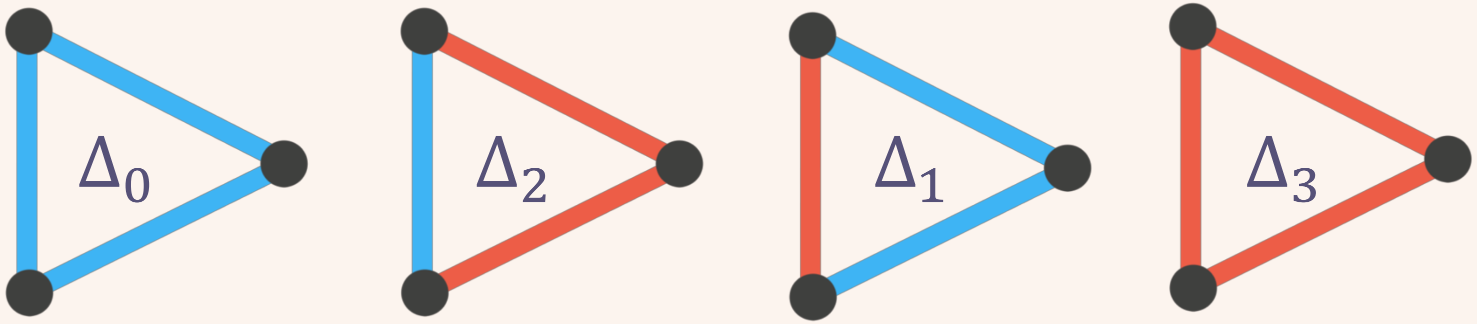

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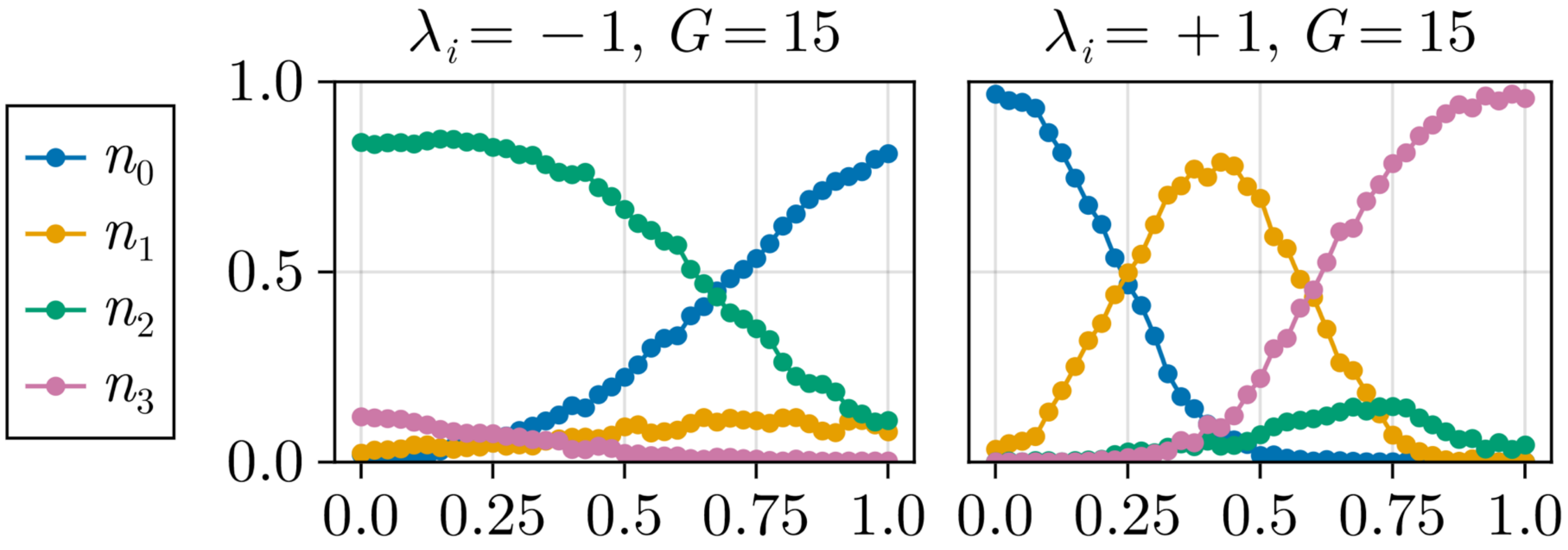

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IT hypotheses

Observation 1: Heterophily minimizes similarities and dissimilarities.
Hypothesis 1: Heterophily reduces low-order dependencies.

Observation 2: Heterophily leads to specific triadic patterns of interaction.
Hypothesis 2: Heterophily promotes high-order dependencies.

For systems (or subsystems) of size $N = 3$:

low-order dependencies: $u_1(\boldsymbol{X}^N) = \frac{1}{3}\sum_{i<j} I\left(X_i; X_j\right)$

high-order dependencies: $u_2(\boldsymbol{X}^N) = \frac{1}{3}\sum_{i<j<k} I(X_i; X_j \mid X_k)$

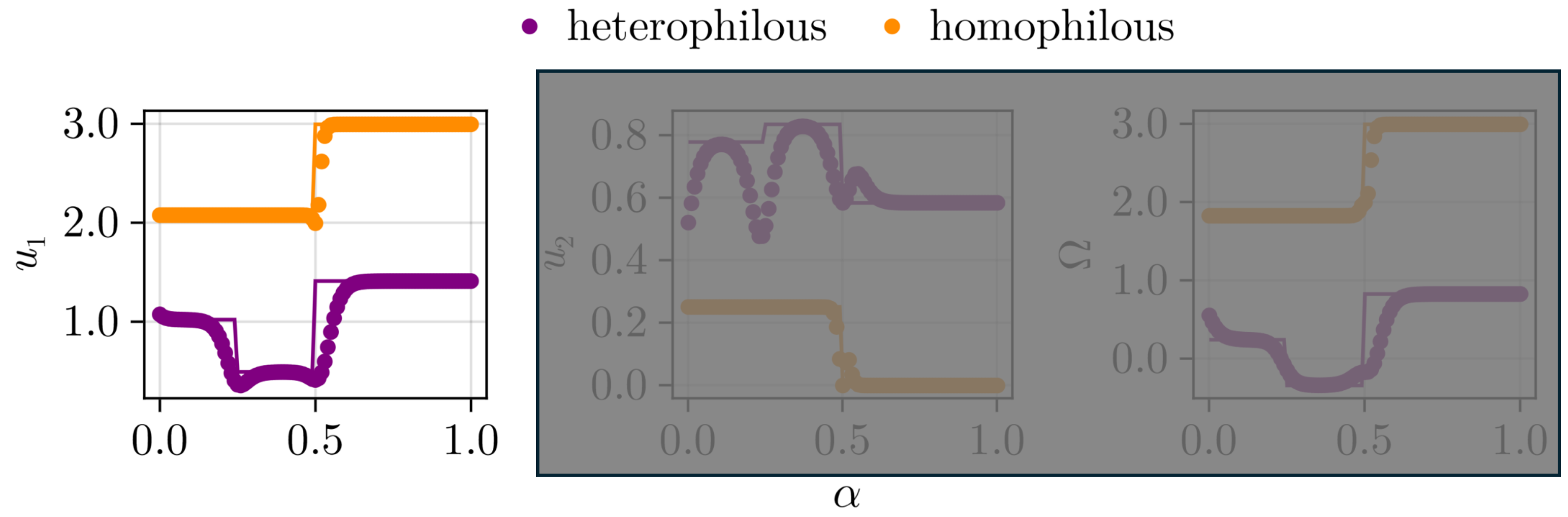
Synergy dominance if: $\Omega(\boldsymbol{X}^N) = u_1(\boldsymbol{X}^N) - u_2(\boldsymbol{X}^N) < 0$

IT results ($G = 3$)

Hypothesis 1: Heterophily reduces low-order dependencies u_1
Hypothesis 2: Heterophily promotes high-order dependencies u_2

Systems size
N=3

Solid line:
analytical results
Markers:
Equilibrium
distribution



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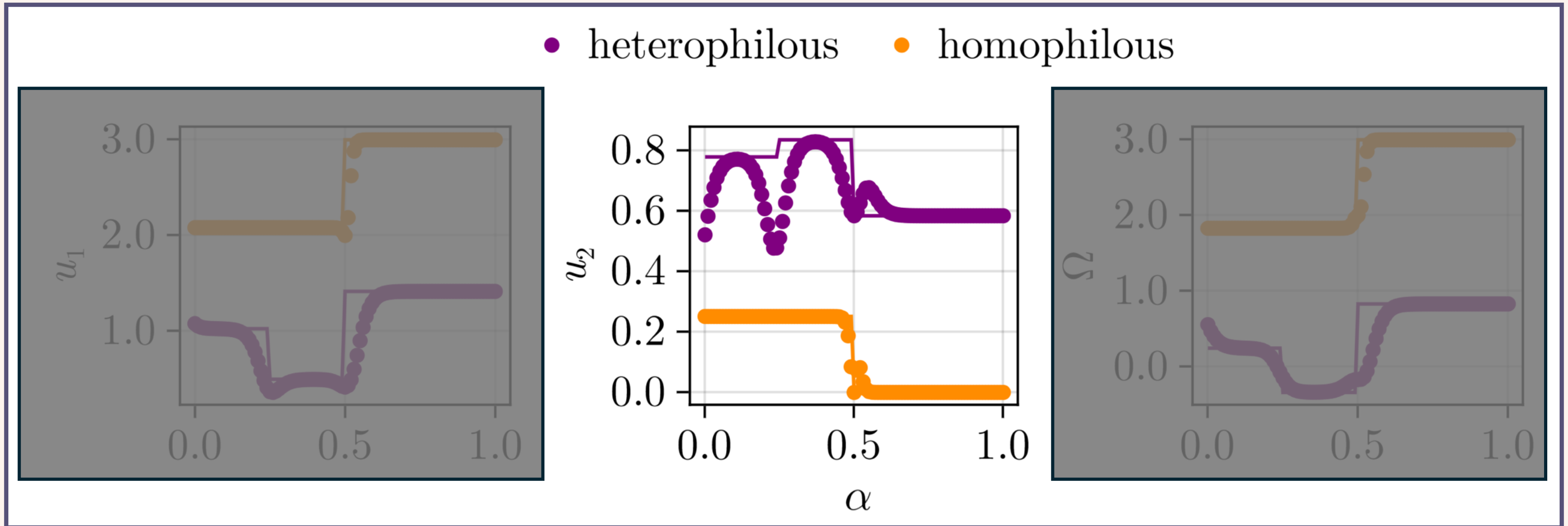

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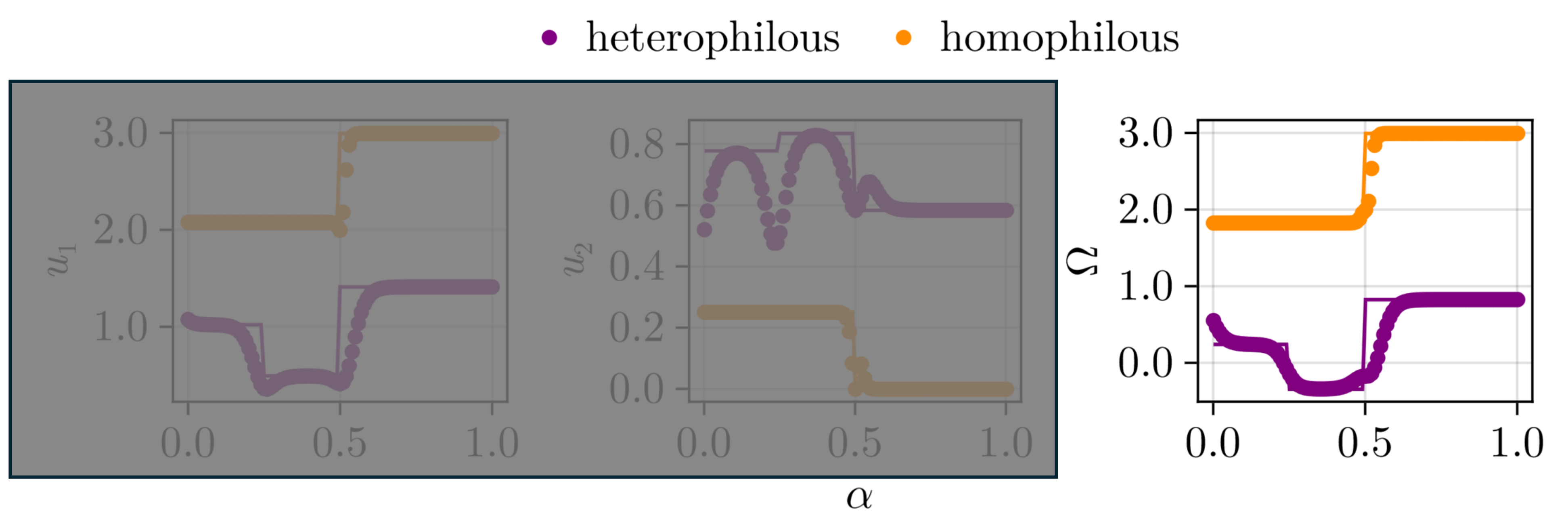

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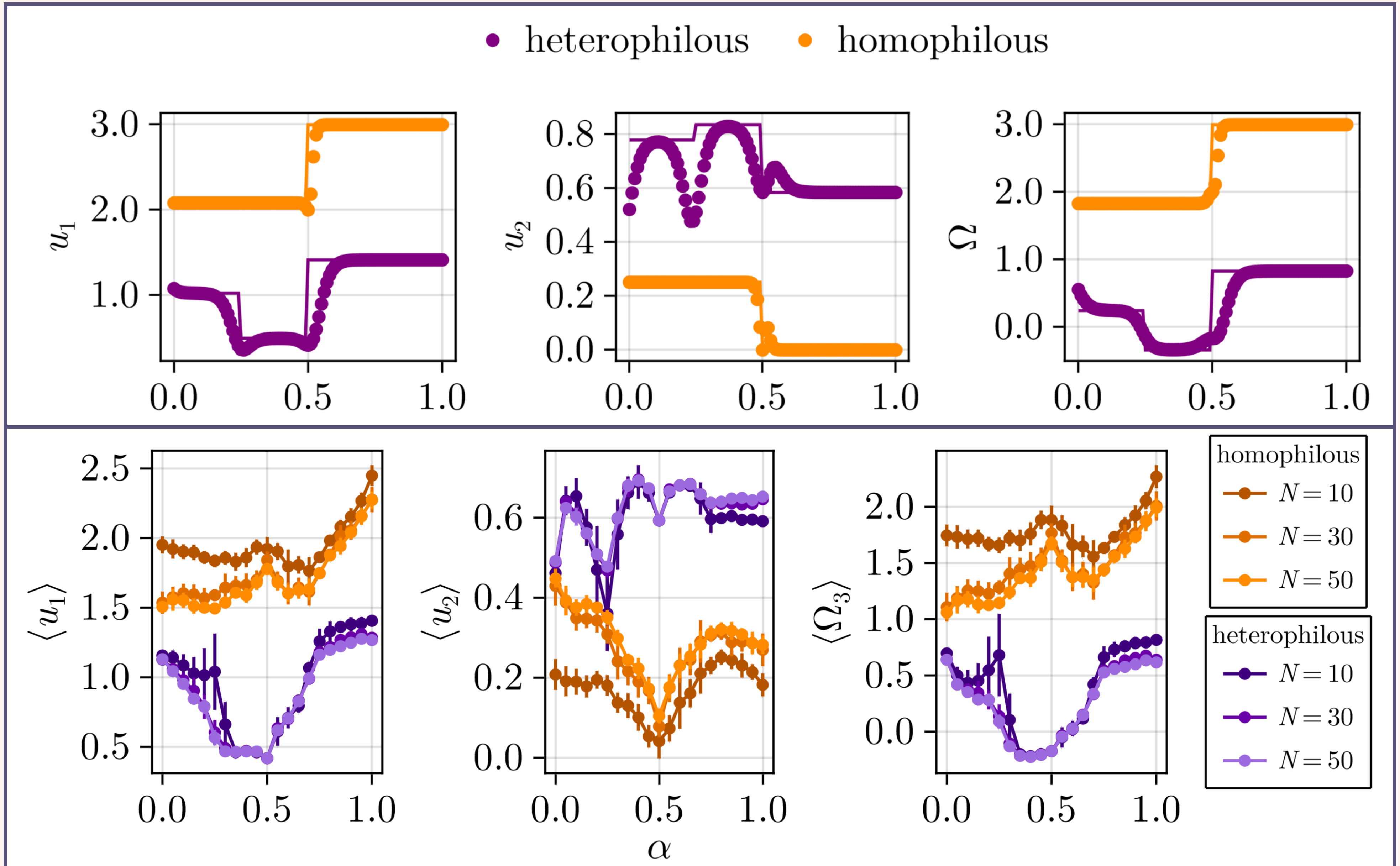
Large Systems (triplet averages)



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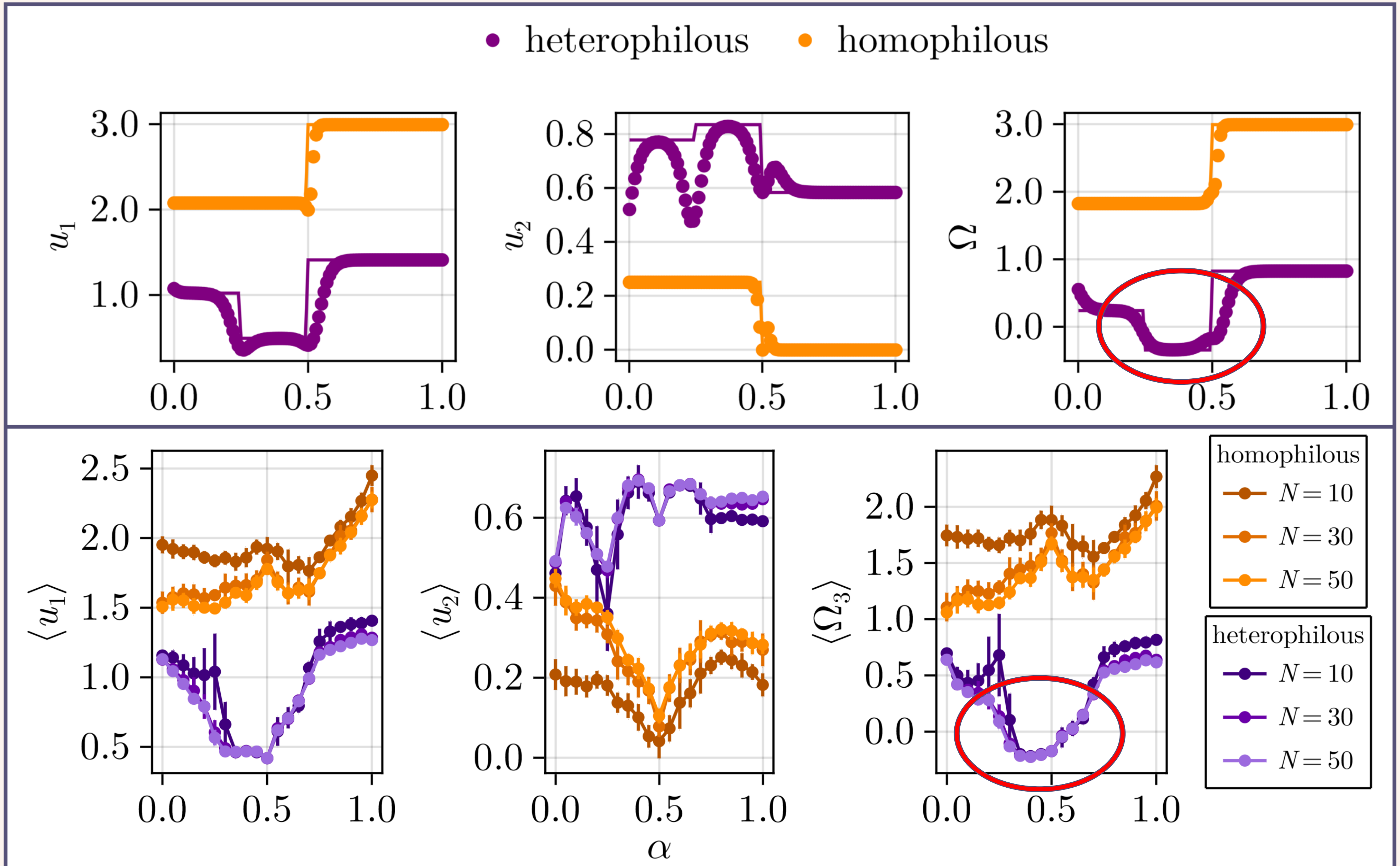
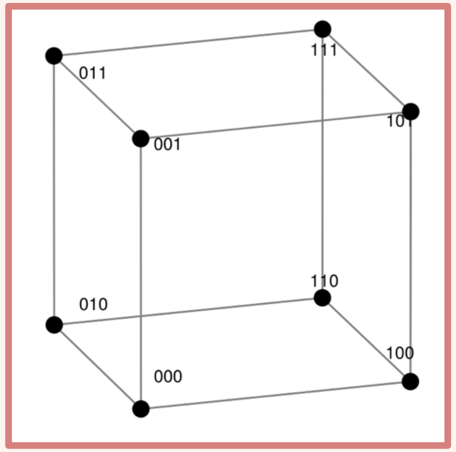
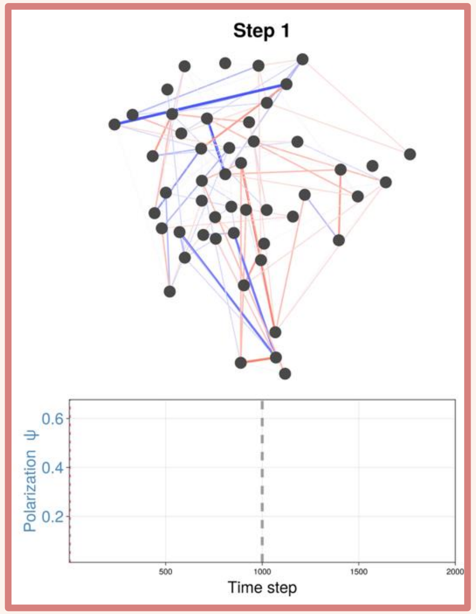


Summary and conclusions

- Heterophily can be a parsimonious route by which synergistic interdependencies emerge.
- **Precise mechanism**: reducing low-order dependencies while selecting particular high-order configurations.
- **Why:** the problem of minimizing all (dis)similarities is a frustrated one (see paper below for the maths!).
- **Case study findings:** heterophily can decrease polarization and induce effective group mechanism for the formation of opinions.
- **Key message:** synergy arises whenever dynamical rules that minimize pairwise dependencies are combined with some global or high-order "constraint".



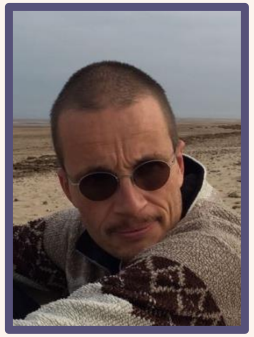


Check out the paper!
"Heterophily as a generative mechanism for self-organized synergistic interdependencies"
ArXiv

Thank you very much and ask me anything!

Limitations?
Simulations?
IT estimations?
Or go have a coffee!

Acknowledgments:



PhD Supervisor
Contributions to all studies

Luc Berthouze

Contributions to "Synergistic Motifs in Gaussian Systems"

Cool quote for the last slide:

"Unbalanced situations stimulate us to further thinking; they have the character of interesting puzzles, problems which make us suspect a depth of interesting background. Sometimes they evoke, like other patterns with unsolved ambiguities, powerful aesthetic forces of a tragic or comic nature." (Heider, 1956)

! ! ! Also, I am looking for a Postdoc ! ! !

I am very open to any **suggestions**, discuss **ideas**, write **proposals**, **you name it**.

Check out the paper!
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Contact details:
Email: ec627@sussex.ac.uk
Bluesky: @enricocaprioglio.bsky.social
Website: enricocaprioglio.github.io/Lucciole

Question and context

Complexity science: systematic investigation of emergent collective phenomena [1]

How and which mechanisms generate collective phenomena?

Collective phenomena: formalized in multivariate-IT as synergistic high-order interdependencies in the patterns of activity of the system.

Mechanisms: entities and activities, as well as their organization, to produce the phenomenon of interest [2].

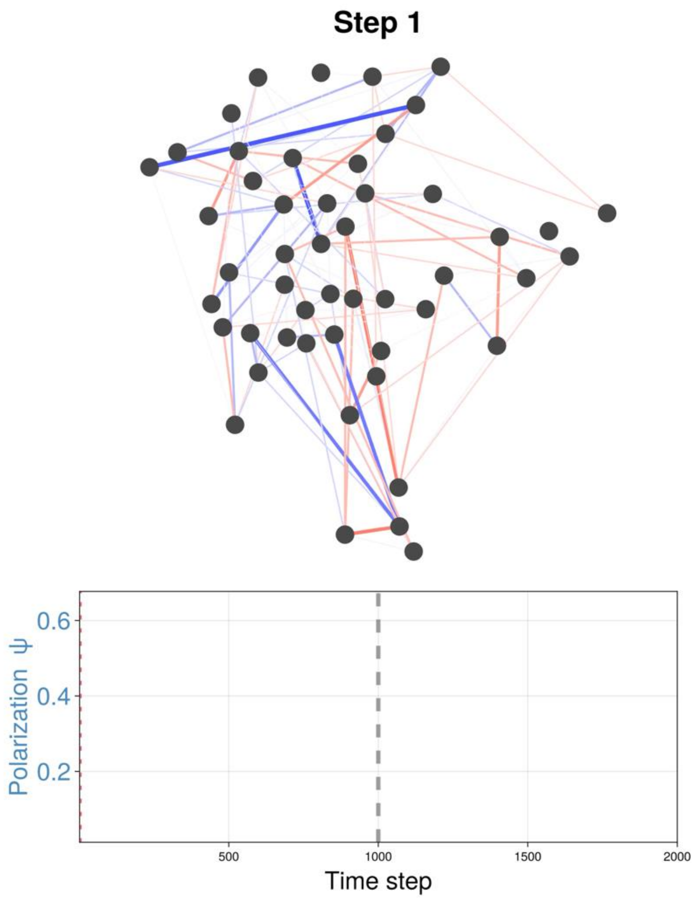
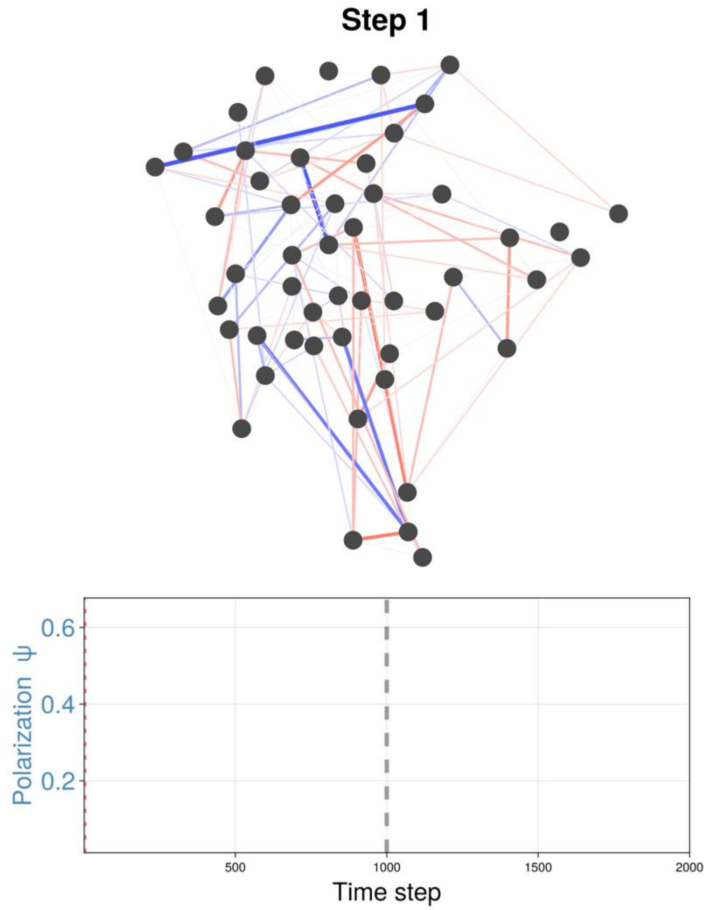
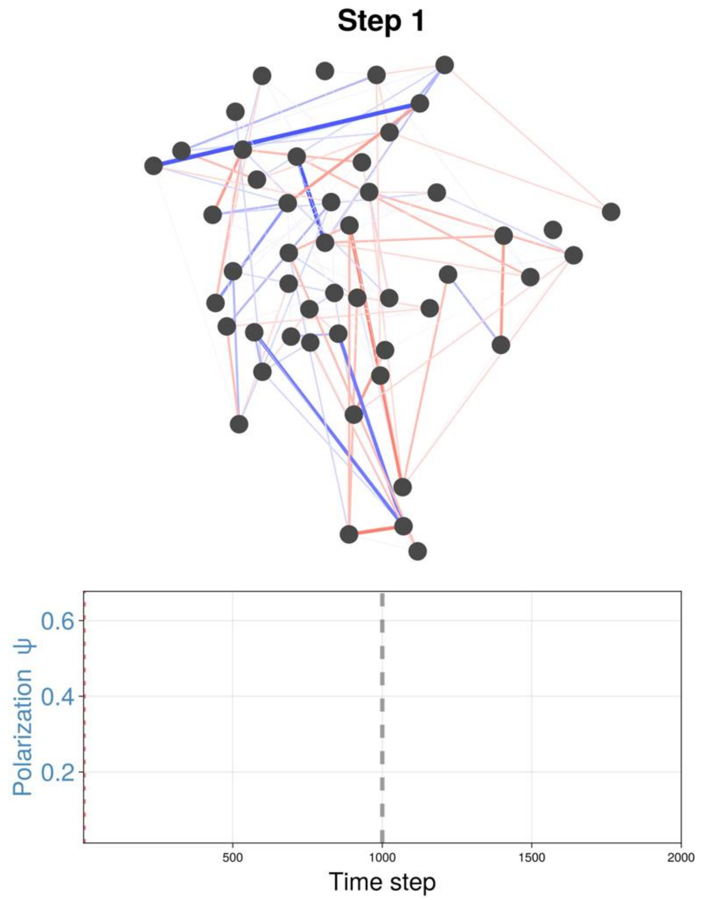
[1] H. J. Jensen, Complexity science (Cambridge University Press, Cambridge, England, 2022)
[2] Craver C, Tabery J, and Illari P. Mechanisms in Science. The Stanford Encyclopedia of Philosophy.

Case study: Disrupting polarization $\psi = \mathrm{var}(s_i \cdot s_j)$

Step 1: simulate polarization (all homophilous see Thurner et al "Why more social interactions lead to more polarization in societies")

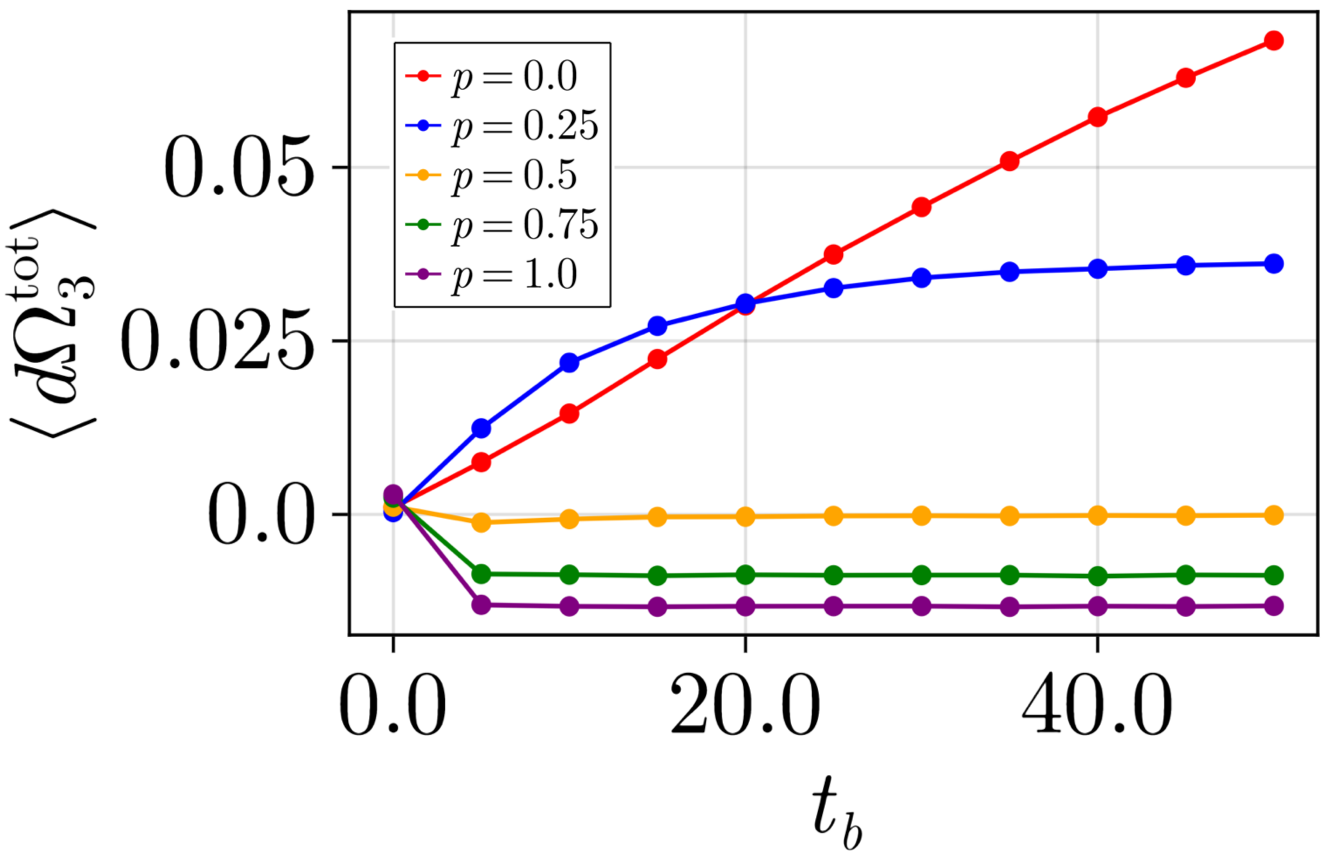
Perturb the system: turn a fraction p of elements from homophilous to heterophilous

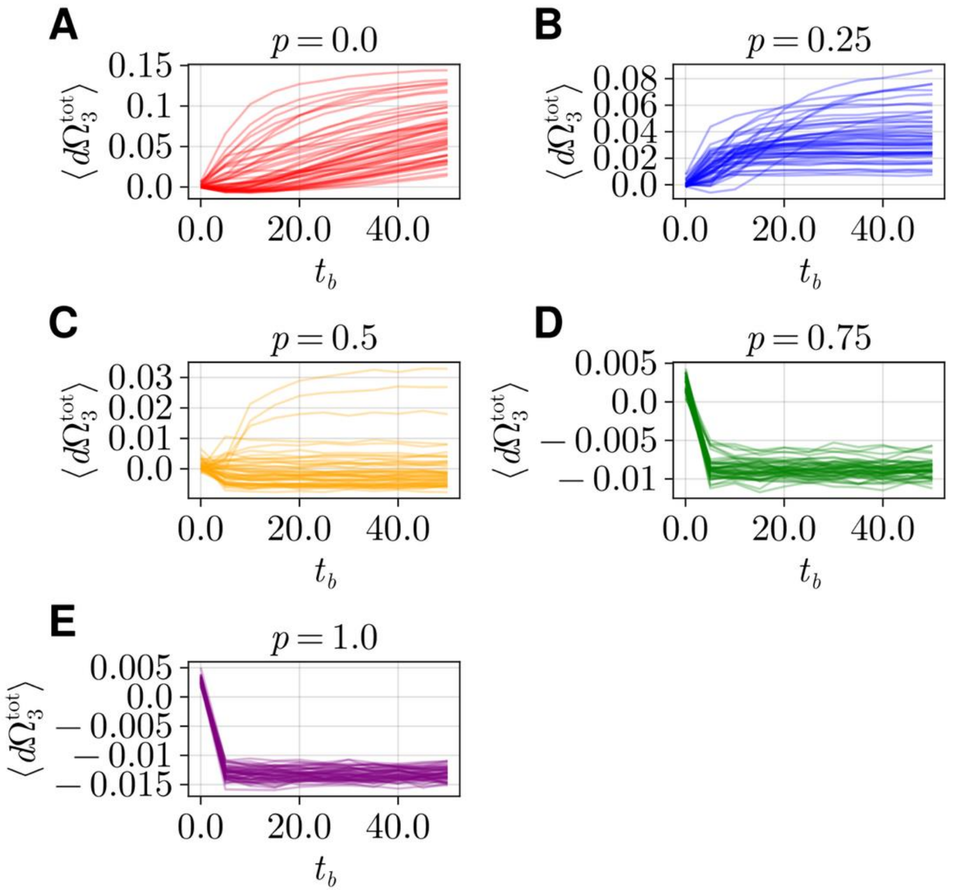
Step 2: continue simulation with heterophilous individuals

$p = 5\%$	$p = 15\%$	$p = 25\%$
		

Case study: Disrupting polarization IT results

Dynamical total O-information, from Robiglio et al. “Synergistic Signatures of Group Mechanisms in Higher-Order Systems” Phys. Rev. Lett.



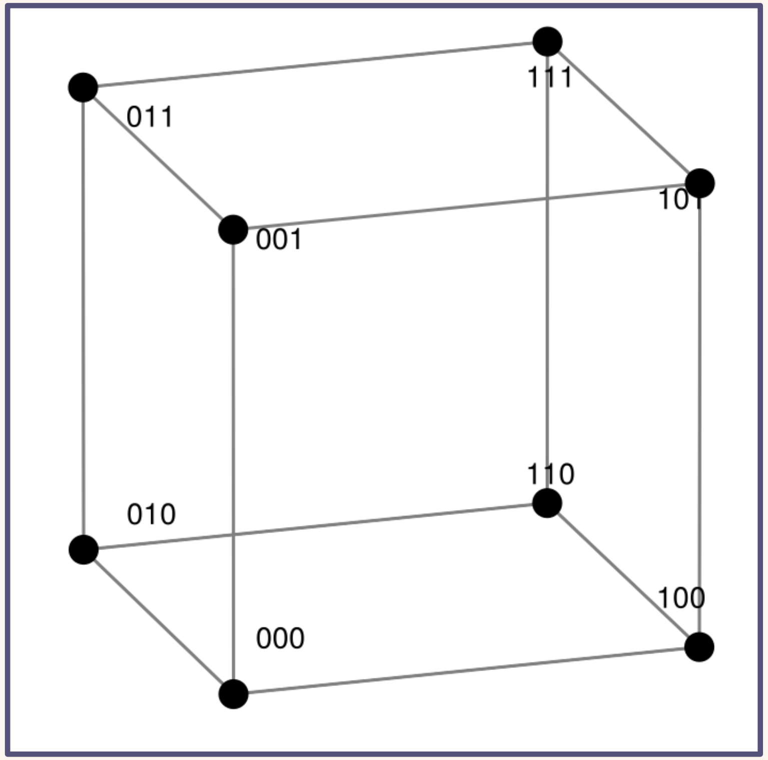


Ground state solutions ($N = 3$)

$\boldsymbol{\lambda}$ pattern	α	$\delta^\star$	Δ_k	E
$(-,-,-)$	$\alpha \leq 1/2$	$\{0, G, G\}$	Δ_2	$-4 + 2\alpha$
	$\alpha \geq 1/2$	$\{0, 0, 0\}$	Δ_0	-6α
$(+,+,+)$ with h odd	$\alpha \leq 1/4$	$\{h-1, h, h\}$	Δ_0	$\frac{10\alpha}{G}$
	$1/4 \leq \alpha \leq 1/2$	$\{h, h, h+1\}$	Δ_1	$\frac{2}{G}(\alpha + 1)$
	$\alpha \geq 1/2$	$\{h+1, h+1, h+1\}$	Δ_3	$\frac{6}{G}(1 - \alpha)$
$(+,+,+)$ with h even	$\alpha \leq 1/2$	$\{h, h, h\}$	Δ_0	$\frac{6\alpha}{G}$
	$1/2 \leq \alpha \leq 3/4$	$\{h, h+1, h+1\}$	Δ_2	$\frac{2}{G}(2 - \alpha)$
	$\alpha \geq 3/4$	$\{h+1, h+1, h+2\}$	Δ_3	$\frac{10}{G}(1 - \alpha)$

TABLE I. For each symmetric λ pattern for systems of size $N = 3$, we report the ground states $\delta^\star$, written in the distance state space notation $\{d_{12}, d_{13}, d_{23}\}$, its associated SBT triangle characterization Δ_k, where k denotes the number of negative edges, and the ground state energy E. For distinct values of α we have different regimes in which a particular configuration $\delta^\star$ is preferred (lowest energy).

More intuitions



Consider the solution (heterophily)

$$\delta = \{d_{12}, d_{13}, d_{23}\}$$
$$\delta = \{1,1,2\}$$

Possible configuration solution:

Start with:

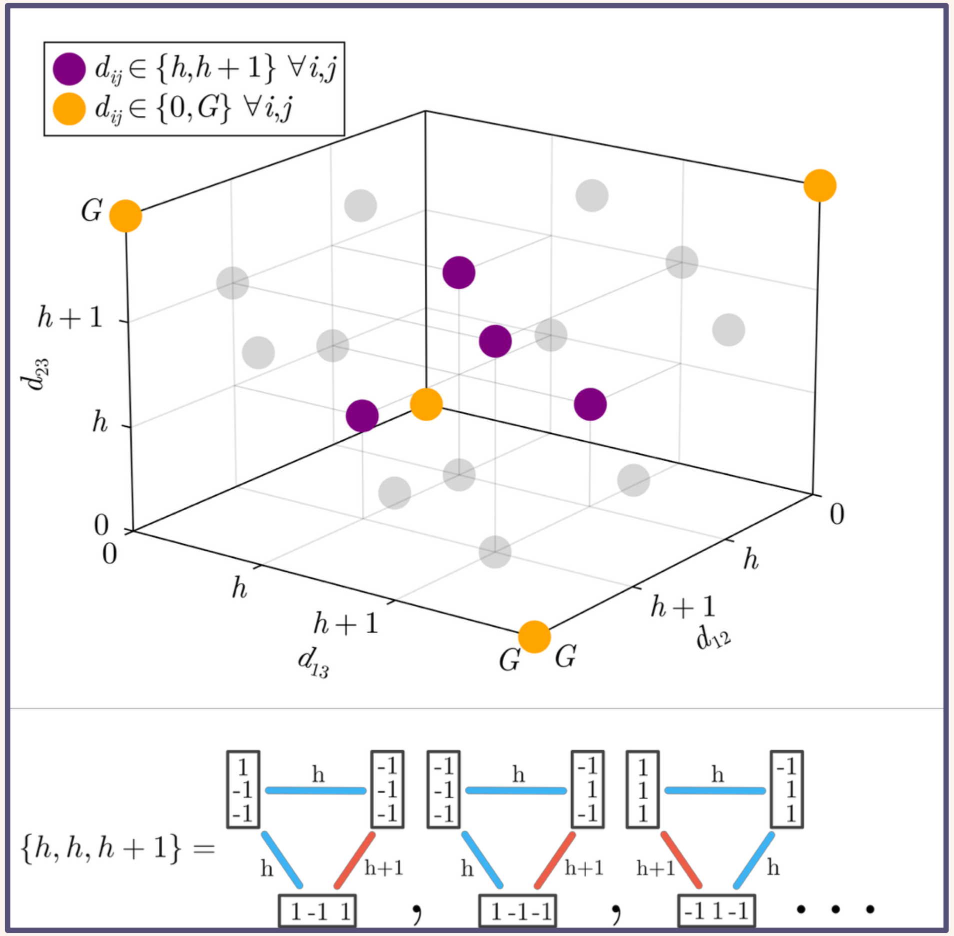
$$s_1 = \{1,1,1\}$$

Flip one spin to obtain

$$s_2 = \{-1,1,1\}$$

Then, to obtain the third spin, we can't flip any random spin again!

Information about the state of all three spins can't be obtained from pairwise information only.

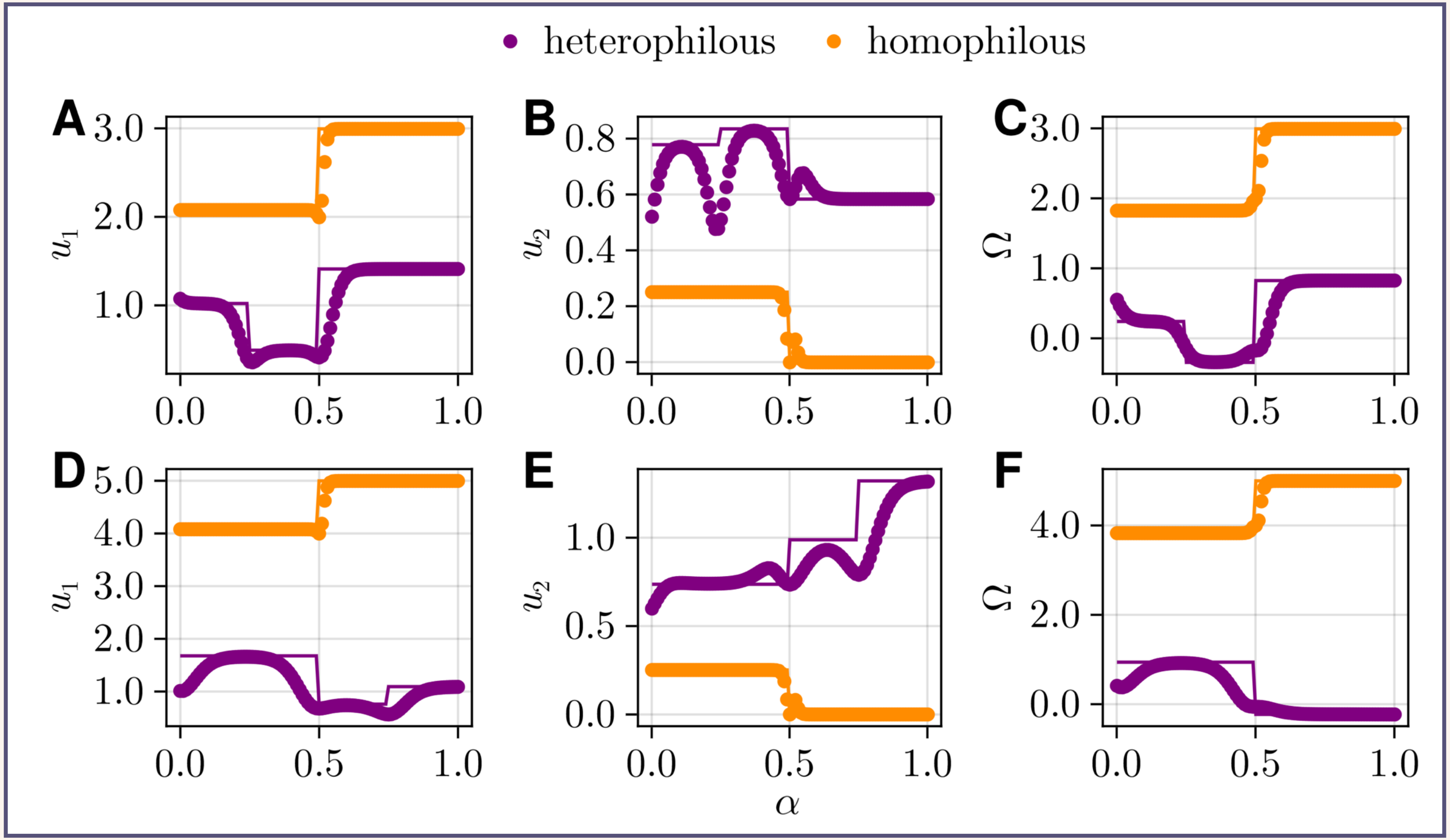


Ground state solutions ($N = 3$)

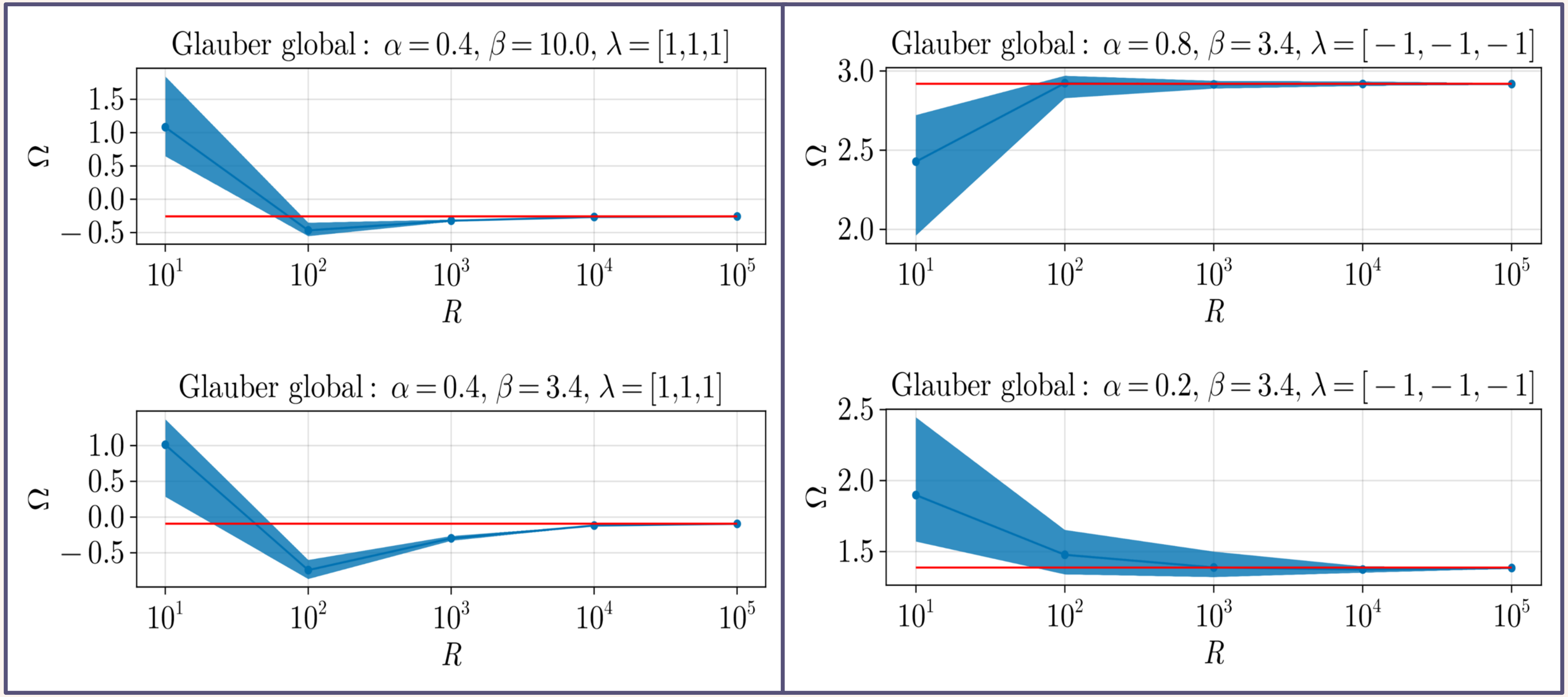
Systems size N=3

Solid line: analytical results

Markers: Equilibrium distribution



Sensitivity analyses



Sensitivity analyses